

HONG KONG INSTITUTE OF INFORMATION TECHNOLOGY
DEPARTMENT OF AI AND BIG DATA

Higher Diploma in Data Science and AI

ITP4475 Big Data and AI Project Management
Project

Group Project

You are required to work individually or in groups of 2 ~ 3. Each individual or group will receive one set of data. All individuals or groups have to process datasets and generate some findings. The data is Vendor Machine Transaction Records. All individuals or groups use the provided data and perform a Crisp-DM approach analysis in order to answer the client's concern. All tasks are required to use R for data import, transformation, mining and fact findings.

Scenario

Background Information

Vending machines combine the concept of online business management with offline convenience services. By adopting a compact, self-service operating model, they significantly reduce labor costs and make affordable, high-quality products readily accessible to consumers. This approach has emerged as another mainstream retail format in recent years. For vending machine operators, key decisions revolve around product replenishment frequency, category selection, supply quantities, and optimal machine placement locations. Scientific business data analysis plays a crucial role here: it helps operators gain deeper insight into user needs, accurately predict product demand, deliver more precise and attentive services, and guide strategic business direction—making it an essential driver for the ongoing development and success of the vending machine retail model.

A Sale Transaction Dataset

In this case, a shopping mall has installed five vending machines at different locations, labelled A, B, C, D, and E. The Sales_Record sheet contains detailed sales data for all products sold through each machine from January 1, 2017, to December 31, 2017, while the Product_Information sheet provides the product categorisation details.

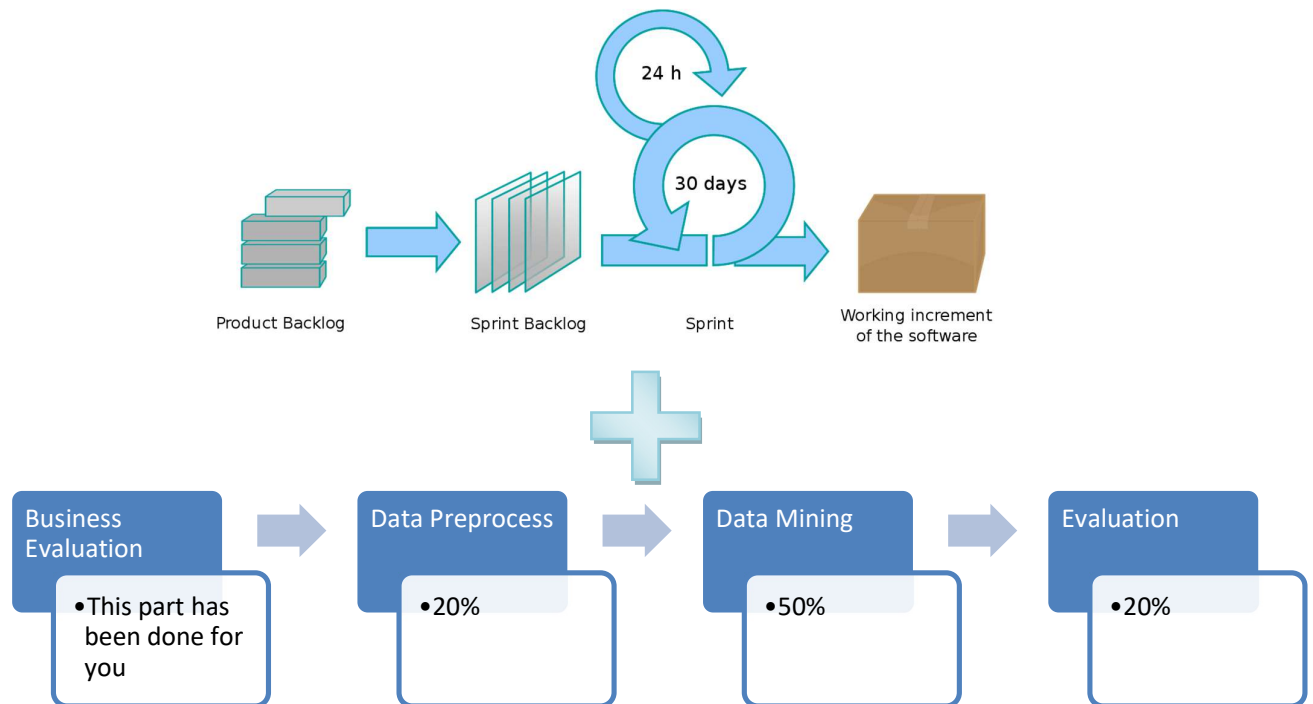
订单号	设备ID	应付金额	实际金额	商品	支付时间	地点	状态	提现
DD201708167493663618499909784	E43A6E078A07631	4.5	4.5	68g好丽友巧克力派2枚	1/1/2017 0:53	D	已出货未退款	已提现
DD201708167493663555814061164	E43A6E078A04172	3	3	40g双汇玉米热狗肠	1/1/2017 1:33	A	已出货未退款	已提现
DD201708167493578526890939886	E43A6E078A06874	5.5	5.5	430g泰奇八宝粥	1/1/2017 8:45	E	已出货未退款	已提现
DD201708167493683507186615837	E43A6E078A04228	5	5	48g好丽友馨愿香烤原味	1/1/2017 9:05	C	已出货未退款	已提现
DD201708167493759548618252006	E43A6E078A04134	3	3	600ml可口可乐	1/1/2017 9:41	B	已出货未退款	已提现
DD2017081016294251D0FA5D314F1	E43A6E078A04134	4.5	4.5	营养快线	1/1/2017 9:41	B	已出货未退款	已提现
DD201708167493663534589050871	E43A6E078A04228	7	7	330ml伊利畅意乳酸菌原味	1/1/2017 10:02	C	已出货未退款	已提现
DD201708167493663526093267894	E43A6E078A04228	8	8	160g盼盼手撕面包	1/1/2017 11:33	C	已出货未退款	已提现

The following is the key data dictionary of the Sale_Data.xlsx

Field (Chinese Field Name)	Type	Description	range	Sample
Order ID (订单号)	String	Unique identification of each transaction	Unique string	
Machine ID (设备 ID)	String	The vendor machine that makes the order	Unique string	
Payable Amount (应付金额)	Double	Total amount of the order that has to be paid	--	4.5
Actual Paid (实际金额)	Double	Actual paid amount	--	4.5
Product (商品)	String	The purchased product	--	營養快線
Location (地点)	String	The Vendor Machine Location	A to E	A
Payment Time(支付时间)	Time Stamp	The DateTime that the payment was made.	In Year 2017	1/1/2017 0:53
Transaction Status (状态)	String	The order status after payment.the term “已出貨未退貨”means the order was done and no refund required.	--	已出貨未退貨
Withdraw (提现)	String	The indicator that the vendor already received the payment. The term “已提现” means the payment already withdrawn by the vendor.	--	已提现
Category (大类)	String	The categorized label of the product	饮料, 非饮料	非饮料
Sub-Category (二级类)	String	The sub-categorized label of the product	--	饼干糕点

Task

Project development approach (Project Management – 10%)



Project Management + Analysis = 100%

Before starting your tasks

Please use **4475project_ForStudent.Rmd** to finish your project. Also, please follow the tip and instructions in the R notebook. Those are all key hints to finish your project.

Project Management (10%)

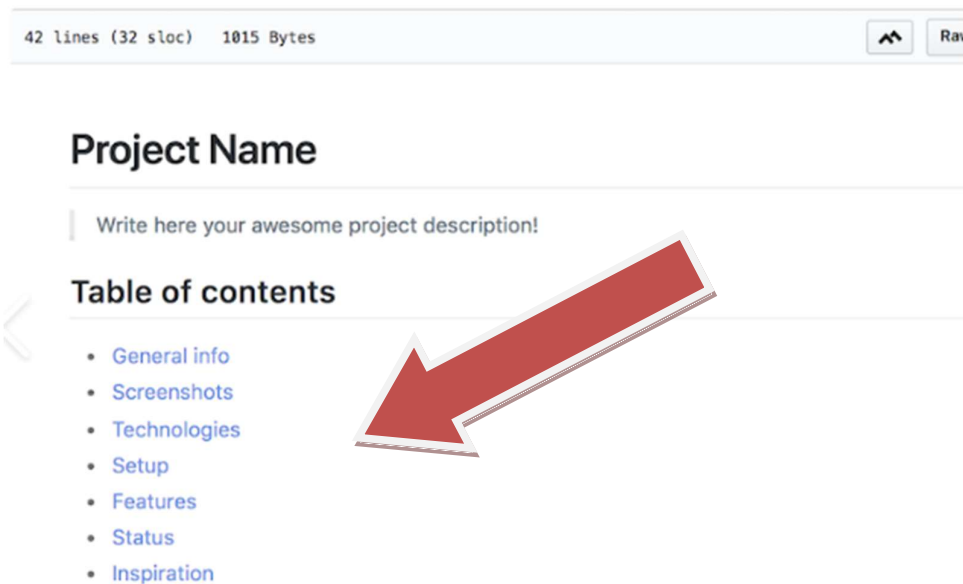
You are required to report all the following items as the project management evidence in the written for the evaluation of your project management. The following criteria should be provided in the **report** document as your Project Management Evidence:

Criteria

1. Private Github Server (2%)
2. Role (2%)
3. Utilisation of GitHub – progress update and readme.md (2%)
4. Product Backlog item records (2%)
5. Sprint Backlog records (2%)

Github Desktop

Your group should nominate a representative to create a private GitHub server and name the repository as “RepresentativeName_4475Project”. Your group have to invite an account bensonlau@vtc.edu.hk as a collaborator. The Repository should include all deliverables, such as an R notebook. You are allowed to create a README .md to announce your project progress.



Scrum Project Development Approach

Your group needs to undergo the Scrum lifecycle during the development of this project. Each teammate should pick one of the roles in scrum and the team has to plan the product backlog and plan the sprint. All this planning and item list should be recorded in the report for the evidence. The document templates are already provided, and you have to fill in all necessary information and place to your written report.

ID	User Story	Tasks	Owner	Status	Estimated effort	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
US001	As a call centre agent I need to be able to see the caller's previous tickets in their contact record.	Install standard contact record module	J Smith	Completed	16	0.5	0	0	0	0	0	0
		Demo standard module and identify revised configuration.	J Smith	Completed		0	8	0	0	0	0	0
		Complete custom configuration as agreed in the demo.	J Smith	Completed		0	0	6	2	0	0	0
US002	As a customer I need to be able to login to my account from any page on the website.	Install login widget on website.	F Dole	Completed	48	0	0	5	7	2	8	6
		Apply customer branding.	J Smith	Completed		0	0	0	0	0	0	0
US003	As a customer I need to be able to look up my address using my postcode.	Install contact page.	P Murphy	Completed	24	0	0	0	0	0	0	0
		Integrate postcode plugin to contact page.	P Murphy	Completed		0	0	0	0	0	0	0
		Demo plugin to customer	J Smith	Completed		0	0	0	0	0	0	0
		Complete any changes required from Demo	J Smith	Completed		0	0	0	0	0	0	0

Business Evaluation

Data has been collected for your action, and you may use the dataset for your project.

Preprocess (20%)

In this part, the mark reward of each requirement contains two components: core code solution and technical explanation. All code should be displayed or written in the provided Rnotebook, and the corresponding code explanation show write in the report document.

应...	实...	商品	支付时间
<chr>	<chr>	<chr>	<chr>
4.5	4.5	68g好丽友巧克力派2枚	42736.03680555556
3	3	40g双汇玉米热狗肠	42736.06458333333
5.5	5.5	430g泰奇八宝粥	42736.364583333336
5	5	48g好丽友薯愿香烤原味	42736.37847222222
3	3	600ml可口可乐	42736.40347222222

The following requirements should be accomplished before modelling:

Requirement	Solution Marks	Explanation Marks
1. Read the dataset and combine two sheets into a single dataframe.	3	2
2. Convert the payment datetime to standard datetime format <pre>Sale\$支付时间 [1] "2017-01-01 00:53:00 UTC" "2017-01-01 01:33:00 UTC" "2017-01-01 08:45:00 UTC" [4] "2017-01-01 09:04:59 UTC" "2017-01-01 09:41:00 UTC" "2017-01-01 09:41:00 UTC" [7] "2017-01-01 10:02:00 UTC" "2017-01-01 11:32:59 UTC" "2017-01-01 11:56:00 UTC" [10] "2017-01-01 11:56:00 UTC" "2017-01-01 12:53:00 UTC" "2017-01-01 12:55:00 UTC"</pre>	2	3
3. Convert all payment related columns to calculable values	2	3
4. Prove that all order ID are unique and there are 70680 unique salse records	2	3
Total:	20	

Data Mining (50%)

In this part, the marks only contain the core code mark. All code should be displayed or written in the provide Rnotebook.

Mission 1 – Data aggregations

Mission 1.1 (1%)

According to the merged dataset, suppose each location only has one vendor machine. Extract the Sales record from the merged dataset and store the data separately based on the Vendor machine location "A", "B", "C", "D", "E".

订单号	设备ID	应付金额	实际金额	商品	支付时间	地点	状态	提现	商品1	大类	二级类
DD20170816749366355814061164	E43A6E078A04172	3.0	3.0	40g双汇玉米热狗肠	2017-01-01 09:33:00	A	已出货未退款	已提现	40g双汇玉米热狗肠	非饮料	肉干/豆制品/蛋
DD201708167493663388737476695	E43A6E078A04172	5.8	5.8	100g卫龙点心里黑椒牛排味	2017-01-02 03:29:00	A	已出货未退款	已提现	100g卫龙点心里黑椒牛排味	非饮料	肉干/豆制品/蛋
DD201708167493663381827746988	E43A6E078A04172	0.8	0.8	顺阳鲜马条 香 西蓝花风味	2017-01-02 03:33:00	A	已出货未退款	已提现	顺阳鲜马条 香 西蓝花风味	非饮料	膨化食品
DD201708167493663503095993712	E43A6E078A04172	3.0	3.0	250ml维他柠檬茶	2017-01-02 03:55:00	A	已出货未退款	已提现	250ml维他柠檬茶	饮料	茶饮料
DD2017081110292488071D623E8F9	E43A6E078A04172	3.5	3.5	东鹏特饮	2017-01-02 04:18:00	A	已出货未退款	已提现	东鹏特饮	饮料	功能饮料
DD201708167493507701935973302	E43A6E078A04172	3.0	3.0	250ml维他柠檬茶	2017-01-02 04:25:00	A	已出货未退款	已提现	250ml维他柠檬茶	饮料	茶饮料
DD2017081018210213E800235B1C8	E43A6E078A04172	3.0	3.0	卫龙大面筋	2017-01-02 04:25:00	A	已出货未退款	已提现	卫龙大面筋	非饮料	肉干/豆制品/蛋
DD2017081110281595114AF30BC0D	E43A6E078A04172	3.5	3.5	东鹏特饮	2017-01-02 16:50:00	A	已出货未退款	已提现	东鹏特饮	饮料	功能饮料
DD201708167493662937167036721	E43A6E078A04172	1.5	1.5	卫龙亲嘴烧红烧牛肉味	2017-01-02 20:02:00	A	已出货未退款	已提现	卫龙亲嘴烧红烧牛肉味	非饮料	肉干/豆制品/蛋
DD201705261437237DD5D9082B98	E43A6E078A04172	4.5	4.5	阿萨姆奶茶	2017-01-02 21:34:00	A	已出货未退款	已提现	阿萨姆奶茶	饮料	茶饮料
DD20170816749300920097772300065	E43A6E078A04172	4.0	4.0	145ml旺仔牛奶盒装	2017-01-02 21:58:00	A	已出货未退款	已提现	145ml旺仔牛奶盒装	饮料	乳制品
DD2017052614334247A1D72F02E30	E43A6E078A04172	2.0	2.0	伊利优酸乳	2017-01-02 23:56:00	A	已出货未退款	已提现	伊利优酸乳	饮料	乳制品
DD201708167493507687358066274	E43A6E078A04172	4.5	4.5	2g辣度海味海苔	2017-01-03 06:42:00	A	已出货未退款	已提现	2g辣度海味海苔	非饮料	海味零食
DD20170811103326649396032790D	E43A6E078A04172	4.0	4.0	脉动	2017-01-03 06:42:00	A	已出货未退款	已提现	脉动	饮料	功能饮料
DD201708167493662547010881037	E43A6E078A04172	3.0	3.0	250ml维他柠檬茶	2017-01-03 17:49:59	A	已出货未退款	已提现	250ml维他柠檬茶	饮料	茶饮料
DD201708167493683259570660391	E43A6E078A04172	2.0	2.0	75g新麦潮天然酵母面包 (北海道牛奶风味)	2017-01-03 22:57:00	A	已出货未退款	已提现	75g新麦潮天然酵母面包 (北海道牛奶风味)	非饮料	饼干糕点

Mission 1.2 (1%)

Calculate each vending machine's sales revenue and number of orders for April 2017, and also compute the overall total sales revenue and total number of orders across all vending machines. Display the results in a table within the report.

Location	Sale_Volume	Revenue
<chr>	<int>	<dbl>
A	440	1780.0
B	601	2462.4
C	742	3251.5
D	443	1687.6
E	887	3723.5

Mission 1.3 (1%)

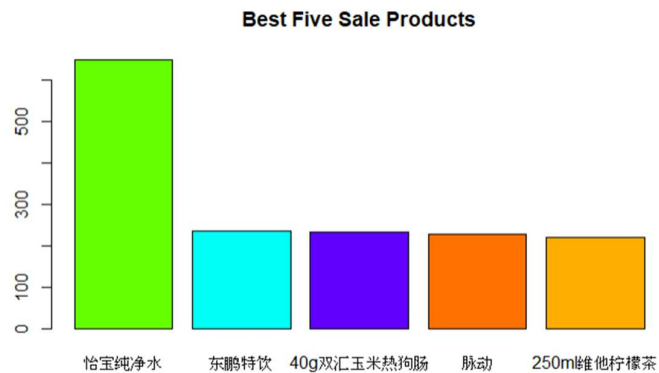
Summarises the monthly average Revenue of each Vendor machine and also the average Sale volume in daily basis.

Location	Month	Revenue	Location	Day	Revenue
<chr>	<chr>	<dbl>	<chr>	<chr>	<dbl>
A	01	4.517485	A	01	343
A	02	3.848837	A	02	427
A	03	3.584337	A	03	285
A	04	4.045455	A	04	323
A	05	4.462159	A	05	409
A	06	4.062608	A	06	326
A	07	4.046521	A	07	337
A	08	3.372687	A	08	296
A	09	4.304126	A	09	355
A	10	4.024269	A	10	352

Mission 2 – Data Modelling and Visualisation

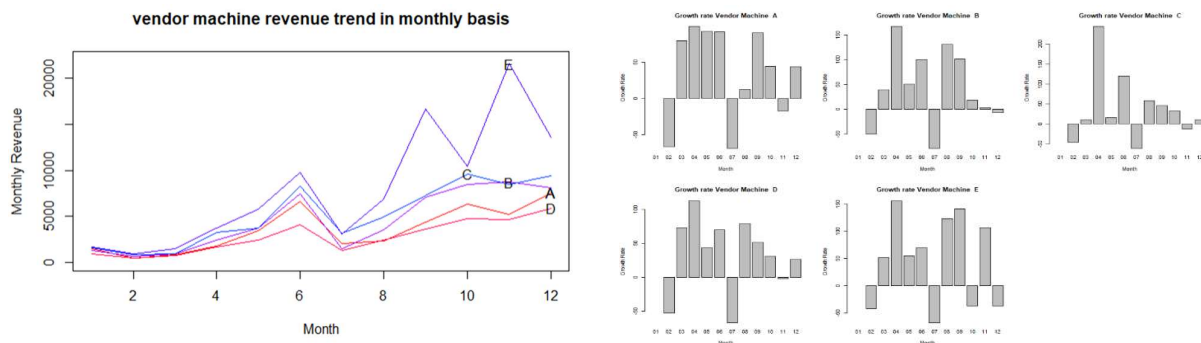
Mission 2.1 (5%)

Visualise the top 5 selling products in June 2017 using a bar chart



Mission 2.2 (6%)

Visualise all vendor machine revenue trend in monthly basis and also its growth rate in a bar chart.



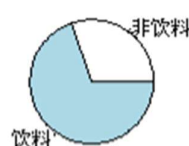
Mission 2.3 (6%)

plot the margin revenue proportion for each vendor machine. Suppose the marginal revenue rate of beverages and non-beverages is 25% and 20%, respectively.

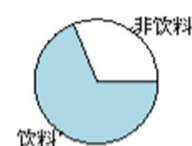
Margin Revenue of Machine A



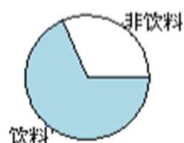
Margin Revenue of Machine B



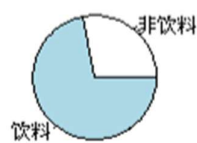
Margin Revenue of Machine C



Margin Revenue of Machine D

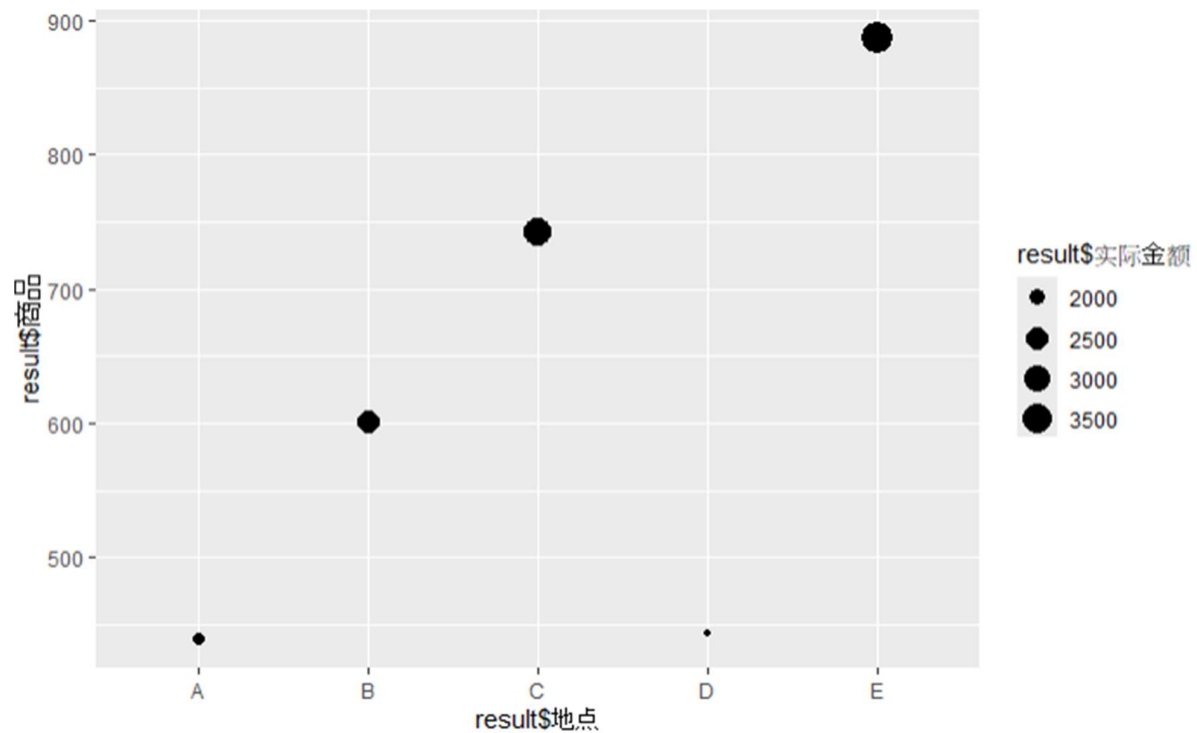


Margin Revenue of Machine E



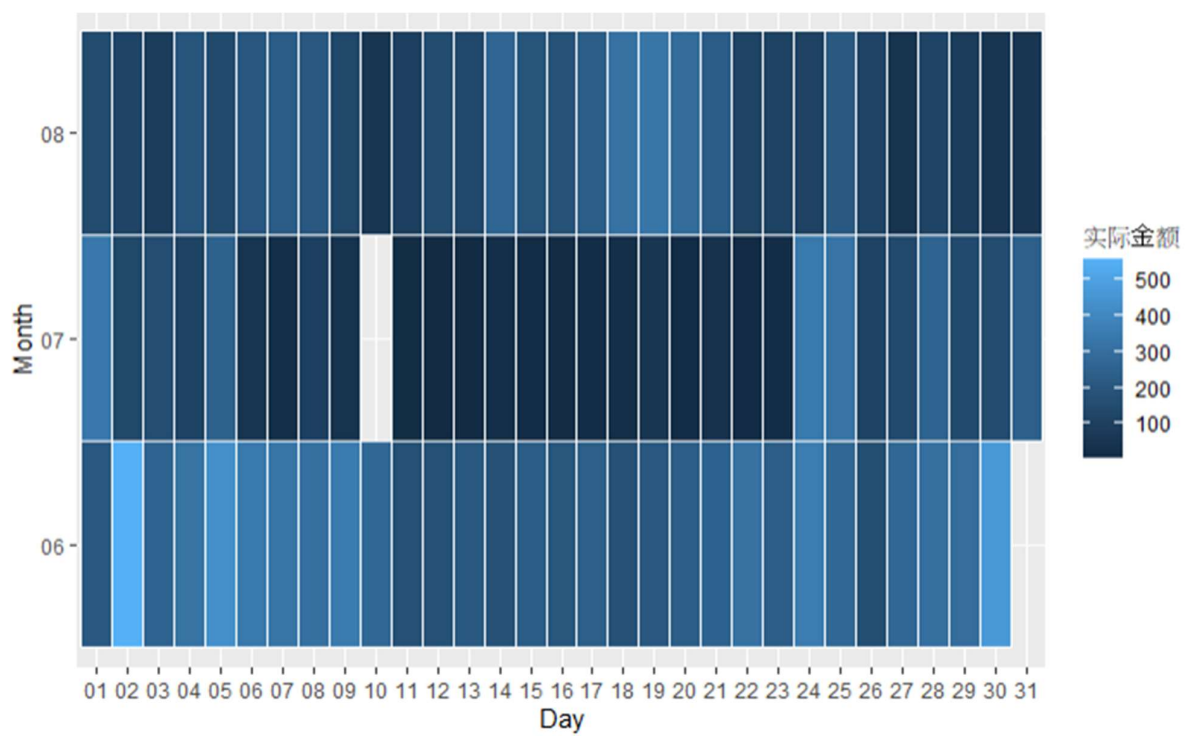
Mission 2.4 (5%)

Plot the total revenue of each machine in April 2027 by bubble chart.



Mission 2.5 (5%)

Plot the revenue of Vendor machine C in June, July, and August and summarise the data by a heatmap.



Mission 3 – Data Mining by Machine Learning Models

Mission 3.1 (10%)

Use any machine learning approach to auto-label the sales status of each beverage product. There are three types of labels: unsaleable, normal, and trending. In which, unsaleable means the sales status was uncertain, and normal means there is no observable sale pattern, trending mean the sale is good.

250ML东鹏特饮	A	unsalable
250ml燕塘原味酸奶	A	normal
250ml燕塘甜牛奶	A	normal
250ml燕塘红枣牛奶	A	normal
250ml王老吉盒装	A	unsalable

Mission 3.2 (10%)

Use any machine learning approach to classify the sales segments of different types of product categories by the relationship of the Category's total Revenue.

Marking Summary

Mission	Code Solution
1.1	1
1.2	1
1.3	1
2.1	5
2.2	6
2.3	6
2.4	5
2.5	5
3.1	10
3.2	10
Total:	50

Evaluation – Answer the following questions from the Vendor. (20%)

1. Provide at least three observations from the Mission 2.5. (5%)
2. Explain the labelling strategy and the corresponding algorithms in Mission 3.1 with details. (5%)
3. Visualise the trending products in Vendor Machine C and provide three observations. (5%)
4. Which category of products will be a trending category in 2018? Explain in detail. (5%)

Reporting

The Report should have the following items.

1. Introduction
2. Data
3. Setting up and Utilising GitHub
4. Roles
5. Product Backlog items
6. Sprint Backlog items
7. Code solution and Explanation
8. Evaluation

Instructions to Students

1. Backup your initial dataset for Data recovery
2. Read the instructions carefully in 4475project_ForStudent.Rmd
3. Check the deliverable carefully before submission:
 - i. 4475_project_ForStudent.Rmd
 - ii. Report.docx